

On a multicriteria shortest path problem

Ernesto Queirós Vieira MARTINS

Universidade de Coimbra, Departamento de Matemática, 3000
Coimbra, Portugal

Received January 1982

Revised January 1983

Multicriteria shortest path problems have not been treated intensively in the specialized literature, despite their potential applications. In fact, a single objective function may not be sufficient to characterize a practical problem completely. For instance, in a road network several parameters (as time, cost, distance, etc.) can be assigned to each arc. Clearly, the shortest path may be too expensive to be used. Nevertheless the decision-maker must be able to choose some solution, possibly not the best for all the criteria.

In this paper we present two algorithms for this problem. One of them is an immediate generalization of the multiple labelling scheme algorithm of Hansen for the bicriteria case. Based on this algorithm, it is proved that any pair of non-dominated paths can be connected by nondominated paths. This result is the support of an algorithm that can be viewed as a variant of the simplex method used in continuous linear multiobjective programming. A small example is presented for both algorithms.

1. Introduction

Multicriteria shortest path problems have not been treated intensively in the specialized literature, despite a reasonable set of potential applications, [5]. In fact, a single objective function may not be sufficient to characterize a practical problem completely. For instance in a road network several parameters (as time, distance, cost, etc.) can be assigned to each arc. Clearly, the shortest path may be too expensive to be used. In this case, the decision-maker must be able to choose a reasonable solution, possibly not the best for all the criteria. This problem is a particular discrete linear multiobjective problem.

The author wishes to thank both referees for their helpful comments and suggestions.

In this paper a generalization for the multicriteria case of the multiple labelling scheme algorithm of Hansen [5] is presented. From this generalization an important connection with a continuous linear multiobjective problem is established. This connection leads us to an algorithm which can be viewed as a variant of the simplex method used in continuous linear multiobjective programming [8,9].

2. Terminology

Let $G(N, A)$ denote a directed network consisting of a finite set $N = \{1, \dots, n\}$ of nodes and a finite set $A \subseteq N \times N$ of m arcs. Each arc is denoted by an ordered pair (i, j) , where $i \in N$ and $j \in N$.

Let $i \in N$ and $j \in N$ be two distinct nodes of $G(N, A)$. We define a path $p_{i,j}$ as a sequence $p_{i,j} = \{i = i_1, (i_1, i_2), i_2, \dots, i_{l-1}, (i_{l-1}, i_l), i_l = j\}$ of alternating nodes and arcs. The sequence will define a chain if the direction of the arcs is unimportant. Obviously a path is also a chain but the converse is not necessarily true. $p_{i,j}$ is called *non elementary path* when some node is repeated in the sequence and is called *elementary path* (or path) when no node is repeated in the sequence. A path (chain) $p_{i,i}$ is called a cycle (circuit), for any $i \in N$.

Let c_{ij}^k , $1 \leq k \leq r$, be r integral values associated with arc (i, j) , for any $(i, j) \in A$. Each c_{ij}^k refers to a cost that must be paid for traversing (i, j) from node $i \in N$ to node $j \in N$. For each k , c_{ij}^k may represent the distance, the cost, the time, etc., associated with the considered arc.

For any $k \in \{1, \dots, r\}$ and for any path p between any pair of nodes, the path cost $c^k(p)$ is defined as $c^k(p) = \sum_{(i,j) \in p} c_{ij}^k$.

A directed and connected network with no circuits will be called a *tree*. A *spanning tree* of $G(N, A)$ is a tree whose set of nodes is N and whose set of arcs is a sub-set of A .

Let s and t be two given distinct nodes of $G(N, A)$ and let $\mathcal{P}$ be the set of all paths from s to t in $G(N, A)$.

Assumption 1. In what follows the existence of at least one path p_s in $G(N, A)$ is assumed for every node $i \in N - \{s\}$.

A directed spanning tree of $G(N, A)$ rooted at s is a spanning tree of $G(N, A)$ where each arc is directed away from s . Let T be some directed spanning tree of $G(N, A)$ rooted at s and let $i \in N$ be some node of $G(N, A)$. It will be said that $i \in N$ is a pendent node of T if $(i, j) \notin T$ holds for every arc $(i, j) \in A$. If i is a pendent node of T , path p_{si} in T is called a branch of T .

In this paper we are concerned with the multicriteria shortest path problem, that is, it is intended to determine a path p_{si} that optimizes $c^k(p_{si})$, for all $k \in \{1, \dots, r\}$. In general the problem has no optimal solution, at least in the normal sense of the meaning of optimal solution. In fact, usually there exists no path that simultaneously optimizes $c^k(p_{si})$, for all $k \in \{1, \dots, r\}$. Nevertheless a privileged set of paths—the set of non-dominated paths or the set of Pareto Optimal Paths—can be determined. This set has an important characteristic: for any Pareto Optimal Path, it is not possible to determine a path which improves some of its r costs without at least one of the remaining ones getting worse.

3. Mathematical formulation

Let us consider the following multicriteria linear problem (P1):

$$\min c^k(x) = \sum_{(i,j) \in A} c_{ij}^k x_{ij} \quad \text{for any } k \in \{1, \dots, r\}, \quad (1)$$

s.t.

$$\sum_j x_{ij} - \sum_j x_{ji} = \begin{cases} 1 & \text{if } i = s, \\ 0 & \text{if } i \neq s, t \ (i = 1, \dots, n), \\ -1 & \text{if } i = t, \end{cases} \quad (2)$$

$$x_{ij} \geq 0 \quad \text{for any } (i, j) \in A. \quad (3)$$

Let X denote the convex polyhedron defined by constraints (2)–(3) and let V denote the set of vertices of X .

Definition 1. Let $x, y \in X$ be two distinct feasible solutions of (P1). x dominates y ($x \text{ D } y$) if and only if $c^k(x) \leq c^k(y)$ holds for all $k \in \{1, \dots, r\}$ and the strict inequality holds at least once.

Definition 2. Let $X_d = \{x \in X \mid \exists y \in X \text{ such that } y \text{ D } x\}$, be the set of dominated solutions of (P1). Then $X_p = X - X_d$ is the set of nondominated solutions of (P1), or the set of Pareto Optimal Solutions of (P1). $V_p = X_p \cap V$ is the set of non-dominated vertices of X .

Assumption 2. From now on it is assumed the existence of some $k \in \{1, \dots, r\}$, for which $c_{ij}^k \geq 0$ holds for any $(i, j) \in A$. With no loss of generality, it is assumed that $k = 1$.

Assumption 3. For any cycle C in $G(N, A)$, it is assumed that $c^k(C) \geq 0$, for all $k \in \{1, \dots, r\}$ and the strict inequality holds for some value of k .

Later on it will be explained why Assumptions 2 and 3 have been made. Let us discretize (P1) adjoining to its constraints the following one:

$$x \in V \quad (4)$$

and let us denote by (P2) the augmented problem.

The following proposition is immediate and stated without proof.

Proposition 1. Every path $p_{si} \in \mathcal{P}$ is a vertex of X and vice-versa. Furthermore, every basic feasible solution of (P1) is a vertex of X .

Proposition 1 makes clear that (P2) is a mathematical formulation for the multicriteria shortest path problem.

A basic feasible solution of (P1) is a spanning tree of $G(N, A)$, [3]. However, (P1) is highly degenerated. As a consequence a vertex of X can be associated with more than one spanning tree of $G(N, A)$. The following proposition plays an important role throughout the paper.

Proposition 2. Every vertex of X is associated with a directed spanning tree of $G(N, A)$ rooted at s .

Proof. Immediate if $x \in V$ is a nondegenerated vertex. So, let us assume that $x \in V$ is a degenerated vertex, that is, $x \in V$ is associated with more than one spanning tree of $G(N, A)$. Let $\mathcal{T}(x)$ be the set of all spanning trees of $G(N, A)$, associated with $x \in V$. In [3] an efficient algorithm is presented for transforming a spanning tree of $G(N, A)$, $T \in \mathcal{T}(x)$, into a directed spanning tree of $G(N, A)$ rooted at s , T' , such that $T' \in \mathcal{T}(x)$.

From Proposition 2 it can be concluded that $\mathcal{P}$ can be completely determined in the set of directed spanning trees of $G(N, A)$ rooted at s . That is, to determine $\mathcal{P}$ it is not necessary to generate the entire set of basic feasible solutions of (P1). From now on, by tree we mean a directed spanning tree of $G(N, A)$ rooted at s .

Clearly, Definitions 1 and 2 can be used for (P2). So, let T_d and T_p denote the set of dominated solutions of (P2) and the set of Pareto Optimal Solutions of (P2), respectively.

Definition 3. $p_{st} \in \mathcal{P}$ is a *nondominated path* if and only if there exists no path $p'_{st} \in \mathcal{P}$ such that $c^k(p'_{st}) \leq c^k(p_{st})$ for all $k \in \{1, \dots, r\}$ and $c^i(p'_{st}) < c^i(p_{st})$ holds for some $i \in \{1, \dots, r\}$.

The following lemma establishes an important difference between (P1) and (P2).

Lemma 1. $V_p \subseteq T_p$.

Proof. For any $T \in V_p$, there exists no $x \in X$ such that $x \text{ D } T$. In particular, there exists no $T' \in V$ ($V \subset X$) such that $T' \text{ D } T$. So, we may conclude that every nondominated vertex of (P1) is a non-dominated solution of (P2). That is, $V_p \subseteq T_p$.

From the proof of Lemma 1, we conclude that $T_p \subseteq V_p$ is not necessarily true. Actually, it is possible for some nondominated solution of (P2), $T \in T_p$, to be dominated by some non basic solution of (P1). This is the case in the example presented in Fig. 1. For the network in this figure, let $x = (x_{12}, x_{13}, x_{14}, x_{24}, x_{34})$. Clearly, $V = \{x^1, x^2, x^3\}$, where $x^1 = (1, 0, 0, 1, 0)$, $x^2 = (0, 0, 1, 0, 0)$ and $x^3 = (0, 1, 0, 0, 1)$, is the set of feasible solutions for (P2). Since $(c^1(x^1), c^2(x^1)) = (0, 20)$, $(c^1(x^2), c^2(x^2)) = (1, 15)$ and $(c^1(x^3), c^2(x^3)) = (2, 0)$ we conclude that $T_p = V$. However, $x^4 = (0.5, 0.5, 0, 0.5, 0.5) \in$

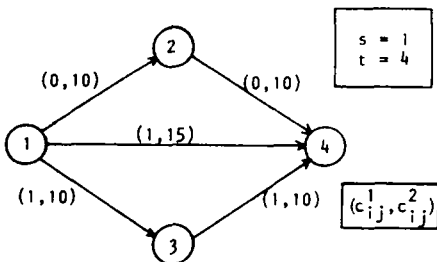


Fig. 1.

X , that is, x^4 is a feasible (non basic) solution of (P1). Since $(c^1(x^4), c^2(x^4)) = (1, 10)$, we conclude that $x^4 \text{ D } x^2$. That is, x^2 is not a Pareto Optimal Solution of (P1).

Some algorithms for determining V_p can be found in the specialized literature [9]. The determination of T_p is not so easy, even if $r = 2$, [5]. Actually the problem has not been treated by many authors; to the best of the author's knowledge only [1,2,5,7] concern this particular discrete multicriteria problem. In [5] Hansen considers ten bicriteria path problems. The complexity of these problems is examined and a multiple labelling scheme is used in the algorithms presented. Also in [7] an algorithm based on a multiple labelling scheme is presented. In [1,2] a bicriteria shortest path algorithm is presented in which a procedure for ranking shortest paths is used. Its generalization for the multicriteria case is briefly discussed in [1].

In this paper we generalize for the multicriteria shortest path problem the algorithms of [5]. Moreover, the relation between V_p and T_p is analysed and an algorithm is derived. This algorithm can be viewed as a variant of a known algorithm for determining V_p , [9], supported by the following theorem

Theorem 1. For every pair T, T' of nondominated vertices of (P1), there exists a sequence $\{T = T_1, T_2, T_3, \dots, T_{l-1}, T_l = T'\}$ of adjacent nondominated vertices of (P1). That is, T and T' are connected by adjacent nondominated vertices of (P1).

Proof. See [8].

In general, Theorem 1 is not valid for the discrete case. For instance, in the network of Fig. 2, only the paths $p_1 = \{1, (1, 2), 2, (2, 3), 3, (3, 4), 4\}$ and $p_2 = \{1, (1, 3), 3, (3, 2), 2, (2, 4), 4\}$ are nondominated. Furthermore, p_1 and p_2 are determined in non adjacent vertices of X . Neverthe-

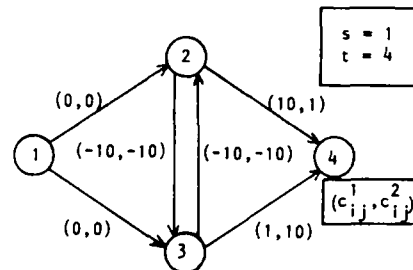


Fig. 2.

less, under Assumption 3, it will be proved that Theorem 1 holds for (P2).

Let us introduce some theoretical consideration.

From now on we abbreviate (sometimes) the notation $p_{s,t}$ to p .

Proposition 3. *Let p_1 , p_2 and p_3 be three distinguished paths from s to t in $G(N, A)$. Then $p_1 \text{ D } p_2$ and $p_2 \text{ D } p_3$ implies that $p_1 \text{ D } p_3$.*

Proof. Immediate.

Lemma 2. *For any dominated path p , there exists a nondominated path p' such that $p' \text{ D } p$.*

Proof. Follows from Proposition 3 and from the fact that the number of paths from s to t in $G(N, A)$ is finite.

Lemma 3. *Under Assumption 3, every non elementary path is dominated by some nondominated path.*

Proof. Let p be a non elementary path. With no loss of generality let us assume that p has a single repeated node i . So, $p = p_{s,i} \cup p_{i,i} \cup p_{i,t}$, where $p_{i,i}$ is a cycle. Then $c^k(p_{i,i}) \geq 0$ for any $k \in \{1, \dots, r\}$ and there exists some $j \in \{1, \dots, r\}$ such that $c^j(p_{i,i}) > 0$. Clearly, $c^k(p) = c^k(p_{s,i}) + c^k(p_{i,i}) + c^k(p_{i,t})$, for any $k \in \{1, \dots, r\}$. Since $c^j(p) > c^j(p_{s,i}) + c^j(p_{i,t})$ and $c^k(p) \geq c^k(p_{s,i}) + c^k(p_{i,t})$ for all $k \neq j$, we may conclude that $(p_{s,i} \cup p_{i,t}) \text{ D } p$. Clearly, $(p_{s,i} \cup p_{i,t})$ is an elementary path from s to t . By Lemma 2, the proof is concluded.

The following lemma was established in [5] for the bicriteria case, without proof.

Lemma 4. *Every nondominated path from s to t in $G(N, A)$, contains only nondominated sub-paths from s to any intermediary node i of the considered path.*

Proof. Let p be a nondominated path from s to t in $G(N, A)$ and let us assume the existence of some intermediary node $i \in p$ such that $p_{s,i}$, the sub-path of p from s to i , is dominated. So, there exists an elementary path $p'_{s,i}$ ($p'_{s,i} \neq p_{s,i}$), such that $c^k(p'_{s,i}) \leq c^k(p_{s,i})$ for any $k \in \{1, \dots, r\}$ and the strict inequality holds at least once. Let $p' = p'_{s,i} \cup p_{i,t}$, where $p_{i,t}$ is the sub-path of p , from i to t . Clearly, $c^k(p') = c^k(p'_{s,i}) + c^k(p_{i,t})$ for any $k \in \{1, \dots, r\}$. So, $c^k(p') \leq c^k(p)$ for any $k \in \{1, \dots, r\}$ and the

strict inequality holds at least once. That is, $p' \text{ D } p$ which is impossible because p is a nondominated path. So we must conclude that p' is a nonelementary path. But if so, from Lemma 3, p' is dominated by some elementary path q . This is also impossible, because from Proposition 3 we would conclude that $q \text{ D } p$.

Lemmas 3 and 4 make clear why Assumption 3 has been made.

Let p_1 and p_2 be two distinguished paths from s to t in $G(N, A)$. It will be said that p_1 and p_2 are adjacent when p_1 and p_2 are determined in two adjacent vertices of X (or in two adjacent trees).

The main result of this paper follows.

Theorem 2. *For every pair p_1, p_2 of nondominated paths, there exists a sequence $\{p_1, p_i, \dots, p_j, p_2\}$ of adjacent nondominated paths. That is, p_1 and p_2 are connected by adjacent nondominated paths from s to t in $G(N, A)$.*

The proof of this theorem is supported by an immediate generalization of the multiple labelling scheme algorithm presented in [5]. This generalization is presented in the next section.

4. Multiple labelling approach

A label is a r -uple of integers and two pointers. During the execution of the algorithm several labels are associated with each node of $G(N, A)$.

Let $i \in N$ be some node of $G(N, A)$. The l th label associated with i is $[(c^1(p_{s,i}), \dots, c^r(p_{s,i})), (j, l_1)]_l$, where j ($j \neq i$) is some node of $G(N, A)$ and l_1 indicates some label of j , for which $c^k_l(p_{s,i}) = c^k_{l_1}(p_{s,j}) + c^k_{j,i}$, for all $k \in \{1, \dots, r\}$. ($c^k_l(p_{s,i})$ is the k th component of the l th label of node i .)

For some node i , we say that $[(c^1(p_{s,i}), \dots, c^r(p_{s,i})), (-, -)]_\xi$ is lexicographically smaller than $[(c^1(p'_{s,i}), \dots, c^r(p'_{s,i})), (-, -)]_\delta$ if $c^1_\xi(p_{s,i}) = c^1_\delta(p'_{s,i}), \dots, c^{k-1}_\xi(p_{s,i}) = c^{k-1}_\delta(p'_{s,i}), c^k_\xi(p_{s,i}) < c^k_\delta(p'_{s,i})$ holds for some $k \in \{1, \dots, r\}$.

We may consider, over the set of all labels associated with each node, the permanent labels and the temporary labels. While the first remain unchanged, the temporary labels can be deleted during the execution of the algorithm. From a permanent label of some node $j \in N$, a temporary label is assigned to every node $i \in N$, such that (j, i) is an arc of $G(N, A)$.

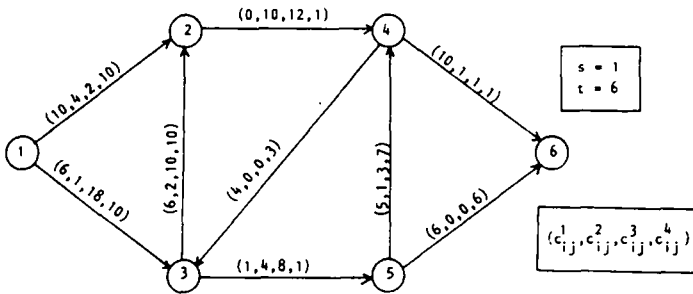


Fig. 3.

Algorithm 1.

Step 0. Assign the temporary label $[(0, 0, \dots, 0), (-, -)]_1$ to node s .

Step 1.

(1) If the set of temporary labels is empty go to step 3. Otherwise, among all the temporary labels determine the lexicographically smallest one. Let it be the l th label associated with node i . Set this label as a permanent one.

(2) While some node $j \in N$ exists, such that $(i, j) \in A$, execute

(1) $c^k(p_{sj}) = c^k_i(p_{si}) + c^k_{ij}$, for every $k \in \{1, \dots, r\}$ and let $[(c^1(p_{sj}), \dots, c^r(p_{sj})), (i, l)]_\xi$ be a new temporary label of node j .

(2) Among all the temporary labels of node j , delete all labels representing a dominated path from s to j .

Step 2. Return to step 1.

Step 3. Find the nondominated paths from s to t . For that, the two pointers of each label must be used.

Step 4. Stop the execution of the algorithm.

Step 3 of Algorithm 1 is easily executed. In fact, at the end of the algorithm there are as many (permanent) labels associated with node $t \in N$ as

the number of nondominated paths from s to t . Moreover, each label corresponds to a unique path. To determine any of these paths, let us choose one label of $t \in N$ and let us assume that (j, l) are the pointers of this label. So, $j \in N$ is the node just before t in the path. To determine the node just before j , we have to know the pointers of the l th label of node j . Clearly, going on backward the first node of the path (s) will be reached.

We must remark that all the nondominated paths from $s \in N$ to all nodes $i \in N - \{s\}$ can be determined with Algorithm 1.

The proof of Algorithm 1 follows clearly from Lemma 4 and from Assumptions 2 and 3.

In order to clarify the stated algorithm, a small example will be solved. So, let us consider the network in Fig. 3. Let us begin to assign the temporary label $[(0, 0, 0, 0), (-, -)]_1$ to node $s = 1$. Clearly, this label will be chosen at the first execution of Step 1 and, as a consequence, is set as a permanent label. From node $s = 1$, only nodes 2 and 3 are temporary labelled with $[(10, 4, 2, 10), (1, 1)]_1$ and $[(6, 1, 18, 10), (1, 1)]_1$, respectively.

In the next execution of Step 1, the first label of node 3, $[(6, 1, 18, 10), (1, 1)]_1$, is chosen as the lexicographically smallest temporary label. From

Table 1

Nodes	Label 1	Label 2	Label 3	Label 4
1	$(0, 0, 0, 0), (-, -)^a$			
2	$(10, 4, 2, 10), (1, 1)^a$	$(12, 3, 28, 20), (3, 1)^a$	$(20, 16, 24, 24), (3, 2)^b$	
3	$(6, 1, 18, 10), (1, 1)^a$	$(14, 14, 14, 14), (4, 2)^a$	$(16, 6, 29, 21), (4, 1)^b$	
4	$(12, 6, 29, 18), (5, 1)^a$	$(10, 14, 14, 11), (2, 1)^a$	$(12, 13, 40, 21), (2, 2)^b$	$(20, 19, 25, 22), (5, 2)^b$
5	$(7, 5, 26, 11), (3, 1)^a$	$(15, 18, 22, 15), (3, 2)^a$		
6	$(13, 5, 26, 17), (5, 1)^a$	$(20, 15, 15, 12), (4, 2)^a$	$(22, 7, 30, 19), (4, 1)^b$	$(21, 18, 22, 21), (5, 2)^b$

^a Permanent labels.

^b Dominated labels.

this label of node 3, only the nodes 2 and 5 are temporary labelled, with $[(12, 3, 28, 20), (3, 1)]_2$ and $[(7, 5, 26, 11), (3, 1)]_1$, respectively.

We must remark that until this moment no temporary label has to be deleted. This will only happen, when from the second label of node 2, $[(12, 3, 28, 20), (3, 1)]_2$, node 4 is labelled with $[(12, 13, 40, 21), (2, 2)]_3$ which is dominated by the first label of node 4.

The procedure is continued until no more temporary labels exists. The final results are reported in Table 1. Clearly, we got two nondominated paths from $s = 1$ to $t = 6$, respectively, $p_1 = \{1, (1, 3), 3, (3, 5), 5, (5, 6), 6\}$ and $p_2 = \{1, (1, 2), 2, (2, 4), 4, (4, 6), 6\}$. For instance, to determine p_2 we must begin with label $[(20, 15, 15, 20), (4, 2)]_2$ of node 6. So, node 6 was labelled from the second label of node 4. The second label of node 4 was obtained from the first label of node 2, which was obtained from the first label of node 1.

5. Proof of Theorem 2

Similarly to the shortest tree, the lexicographically shortest tree is a tree in which a lexicographically shortest path from s to i is determined, for any node $i \in N - \{s\}$.

We will say that a tree is nondominated if all its branches are nondominated paths.

Proposition 4. *The lexicographically shortest tree is a nondominated tree.*

Proof. Let us assume that T , the lexicographically shortest tree, is dominated. So, there exists $i \in N$, a pendent node of T , for which $p_{s,i}$ is dominated ($p_{s,i}$ is determined in T). That is, there exists a path $p'_{s,i}$ such that $c^k(p'_{s,i}) \leq c^k(p_{s,i})$ holds for any $k \in \{1, \dots, r\}$ and $c^j(p'_{s,i}) < c^j(p_{s,i})$ holds for some $j \in \{1, \dots, r\}$. So, T is not the lexicographically shortest tree which contradicts the assumption.

We must remark that the lexicographically shortest tree is easily determined using any of the available algorithms for the shortest tree problem [6]. Clearly a slight and obvious modification has to be introduced in such algorithms. The lexicographically shortest tree can also be determined with Algorithm 1. Clearly, after the execution of

Algorithm 1, the lexicographically smallest label of each node $i \in N - \{s\}$ is the value of the lexicographically shortest distance from s to i . From these labels, the lexicographically shortest tree can be easily constructed, because every lexicographically shortest path is formed by lexicographically shortest sub-paths.

Let us consider now the second lexicographically smallest label of node t , which is the value of the distance of the second lexicographically shortest path from s to t . Only two cases can be considered. This label was obtained either from a lexicographically smallest label or from a second lexicographically smallest label. In this case, we also started with either a lexicographically smallest label or with a second lexicographically smallest label. As node s has only a single label, going on backward, a lexicographically smallest label will be reached. So, the second lexicographically shortest path from s to t is determined in a nondominated tree, adjacent of the lexicographically shortest tree.

With a similar reasoning, the following lemma could be concluded.

Lemma 5. *Let $p_{s,i}$ be some nondominated path but not the lexicographically shortest one, unless the lexicographically shortest path is not unique. There exists a nondominated path $p'_{s,i}$, adjacent to $p_{s,i}$, such that one of the following cases holds:*

- (1) $c^k(p'_{s,i}) = c^k(p_{s,i})$ for any $k \in \{1, \dots, r\}$.
- (2) $[c^1(p'_{s,i}), \dots, c^r(p'_{s,i})] <^L [c^1(p_{s,i}), \dots, c^r(p_{s,i})]$, where $<^L$ means lexicographically smaller.

The proof of Theorem 2 follows easily from Lemma 5.

6. A basic approach

Based on Theorem 2 a basic algorithm can be stated. In fact, from a nondominated path we only have to generate all its adjacent paths. The lexicographic order will be used to generate nondominated paths, starting from the lexicographically shortest one.

The algorithm is said to be basic, because it only uses bases of (P1).

Let T be some tree (so, T is a basis of (P1)). Associated with T , let $\pi_i^k(T)$ —or simply π_i^k —be the dual variable associated with node i , for any

$i \in N$, when we consider $c^k(x) = \sum c_{ij}^k x_{ij}$ as the objective function. From now on, let $\pi_s^k(T) = 0$, for any $k \in \{1, \dots, r\}$ and for any tree T . Clearly, $\pi_i^k(T)$ is the value of $c^k(p_{si})$, where p_{si} is the path from s to i in T .

For any $k \in \{1, \dots, r\}$ and for arc $(i, j) \in A$, the reduced cost $\bar{c}_{ij}^k(T)$ —or simply $\bar{c}_{ij}^k$ —is defined as $\bar{c}_{ij}^k = \pi_i^k - \pi_j^k + c_{ij}^k$.

Let T be some tree and let $(i, j) \notin T$ be some nonbasic arc. Then $T' = T \cup \{(i, j)\} - \{(h, j)\}$ where $(h, j) \in T$, is an adjacent tree of T . Moreover $c^k(p'_{si}) = c^k(p_{si}) + x_{ij}\bar{c}_{ij}^k(T)$, for any $k \in \{1, \dots, r\}$ and for any $i \in N$, where $x_{ij} = 1$ if and only if $(i, j) \in p_{si}$, and $x_{ij} = 0$, otherwise.

For more details about this subject see [6].

In the algorithm a vector $v \neq 0$ is called lexicographically positive if its first nonzero component is positive. We use the notation $v >^L 0$ to denote v as a lexicographically positive vector.

Algorithm 2.

Step 0.

(1) Determine T , a lexicographically shortest tree.

(2) Compute $[\pi_i^1(T), \dots, \pi_i^r(T)]$ for any $i \in N$.

(3) Place T in LIST.

Step 1. If LIST is empty then finish the algorithm. Otherwise remove from LIST a tree T' with lexicographically smallest $[\pi_i^1(T'), \dots, \pi_i^r(T')]$. Compute $[\bar{c}_{ij}^1(T'), \dots, \bar{c}_{ij}^r(T')]$ for any $(i, j) \notin T'$.

Step 2. If p' (the path from s to t in T') does not belong to LIST1, then place p' in LIST1.

Step 3. While there exists $(i, j) \notin T'$ such that either $[\bar{c}_{ij}^1(T'), \dots, \bar{c}_{ij}^r(T')] >^L 0$ and $\bar{c}_{ij}^k < 0$ holds for some

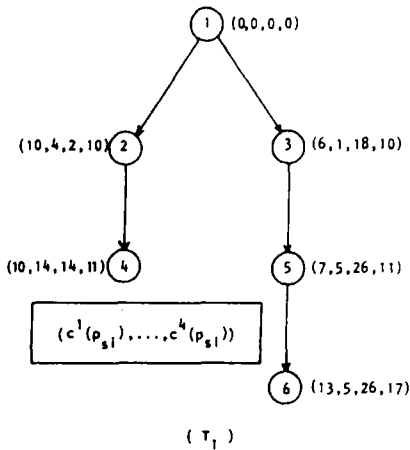


Fig. 4.

Table 2

$(i, j) \notin T_1$	$\bar{c}_{ij}^1$	$\bar{c}_{ij}^2$	$\bar{c}_{ij}^3$	$\bar{c}_{ij}^4$
(3, 2)	2	-1	26	10
(4, 3)	8	13	-4	4
(4, 6)	7	10	-11	-5
(5, 4)	2	-8	15	7

$k \in \{1, \dots, r\}$, or $\bar{c}_{ij}^k = 0$ holds for all $k \in \{1, \dots, r\}$, execute

(1) $T = T' \cup \{(i, j)\} - \{(h, j)\}$, where $(h, j) \in T'$.

(2) Compute $[\pi_i^1(T), \dots, \pi_i^r(T)]$ for all $i \in N$.

(3) If T does not belongs to LIST, then place T in LIST.

Step 4. Delete from LIST all dominated trees.

Step 5. Return to Step 1.

We must remark that in Step 3(1), $T' \cup \{(i, j)\}$ never contains a cycle. If so, this implies that p' (the path from s to t in T') is a nonelementary path and from Lemma 3 we will conclude that $\bar{c}_{ij}^k \geq 0$ for all $k \in \{1, \dots, r\}$.

We will use Algorithm 2 for solving the same example that was solved with Algorithm 1.

The lexicographically shortest tree T_1 , is that one of Fig. 4.

So, T_1 is removed from LIST in the first execution of Step 1. The reduced costs $\bar{c}_{ij}^k(T_1)$ ($k = 1, \dots, 4$) are computed (see Table 2).

Then $p_1 = \{1, (1, 3), 3, (3, 5), 5, (5, 6), 6\}$ and $\text{LIST} = \{p_1\}$. In the first execution of Step 3, trees T_2, T_3, T_4 and T_5 (Fig. 5) are generated and placed in LIST. So, $\text{LIST} = \{T_2, T_3, T_4, T_5\}$.

In Step 4, T_2 and T_3 are removed from LIST, because $T_5 \text{ D } T_2$ and $T_4 \text{ D } T_3$. So, $\text{LIST} = \{T_4, T_5\}$. Then (Step 1) T_5 is removed from LIST and the reduced costs $\bar{c}_{ij}^k(T_5)$ ($k = 1, \dots, 4$) are computed (Table 3).

Since $p_5 = p_1$, Step 3 is then executed to generate a unique tree T_6 (Fig. 6). Then, after the execution of Step 4, $\text{LIST} = \{T_4, T_6\}$.

Table 3

$(i, j) \notin T_5$	$\bar{c}_{ij}^1$	$\bar{c}_{ij}^2$	$\bar{c}_{ij}^3$	$\bar{c}_{ij}^4$
(3, 2)	2	-1	26	10
(4, 3)	10	5	11	11
(4, 6)	9	2	4	2
(2, 4)	-2	8	-15	-7

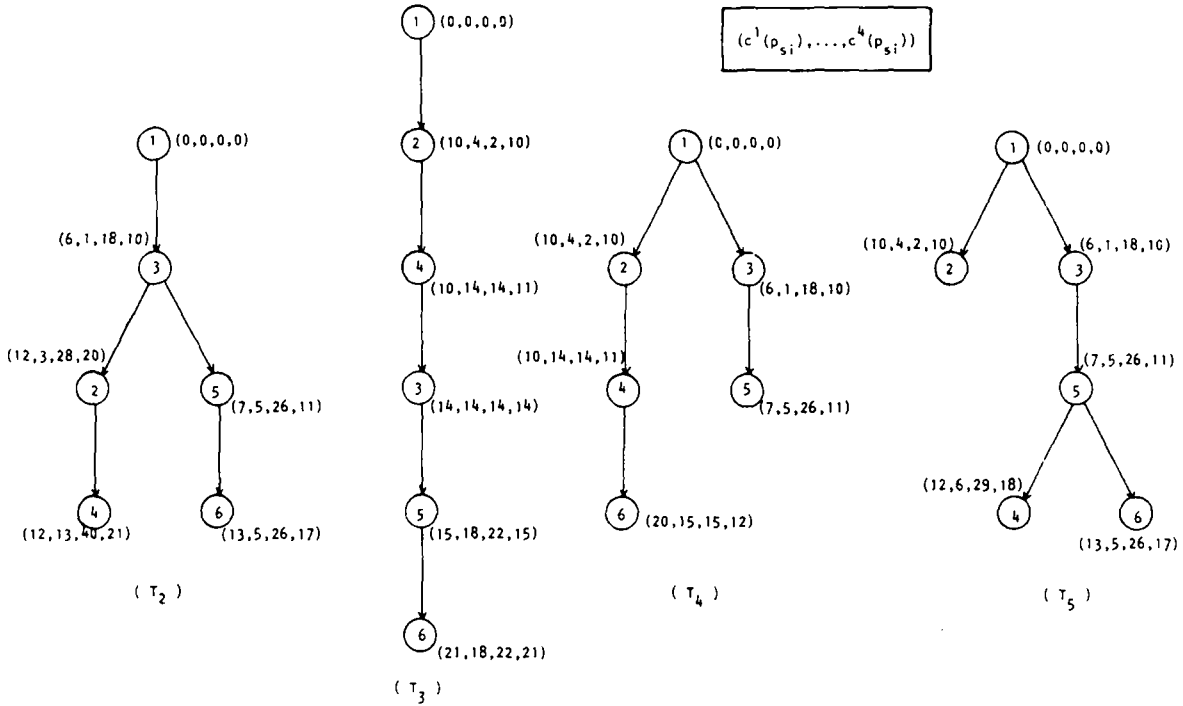


Fig. 5.

In Step 1, T_6 is removed from LIST and the reduced costs $\bar{c}_{ij}^k(T_6)$ ($k = 1, \dots, 4$) are computed (Table 4).

Since $p_6 = p_1$, Step 3 is executed and no tree is generated. So, after the execution of Step 4, LIST = $\{T_4\}$. In Step 1, T_4 is removed from LIST and the reduced costs $\bar{c}_{ij}^k(T_4)$ ($k = 1, \dots, 4$) are computed (Table 5).

Then (Step 2), $p_4 = \{1, (1, 2), 2, (2, 4), 4, (4, 6), 6\}$ and LIST1 = $\{p_1, p_4\}$. In Step 3, trees T_7 , T_8 and T_9 are generated (Fig. 7) and placed in LIST. But $T_6 \text{ D } T_7$ and $T_5 \text{ D } T_9$ (Step 4), so after the execution of Step 4, LIST = $\{T_8\}$.

In Step 1, T_8 is removed from LIST and the reduced costs $\bar{c}_{ij}^k(T_8)$ ($k = 1, \dots, 4$) are computed (Table 6).

In Step 3 no tree is generated and LIST is

empty. So, Algorithm 2 finish with p_1 and p_4 as the nondominated paths from $s = 1$ to $t = 6$.

7. Proof of Algorithm 2

let us assume that at the end of Algorithm 2 there exists some nondominated path $p_{s'}$ that has

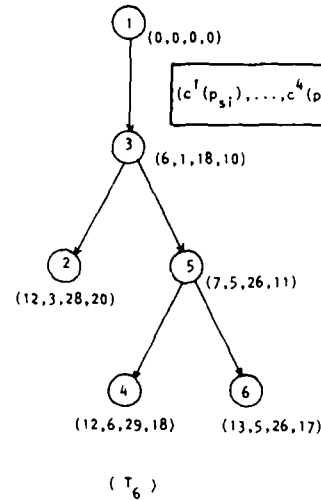


Fig. 6.

Table 4

$(i, j) \in T_6$	$\bar{c}_{ij}^1$	$\bar{c}_{ij}^2$	$\bar{c}_{ij}^3$	$\bar{c}_{ij}^4$
(1, 2)	-2	1	-26	-10
(4, 3)	10	5	11	11
(4, 6)	9	2	4	2
(2, 4)	0	7	11	3

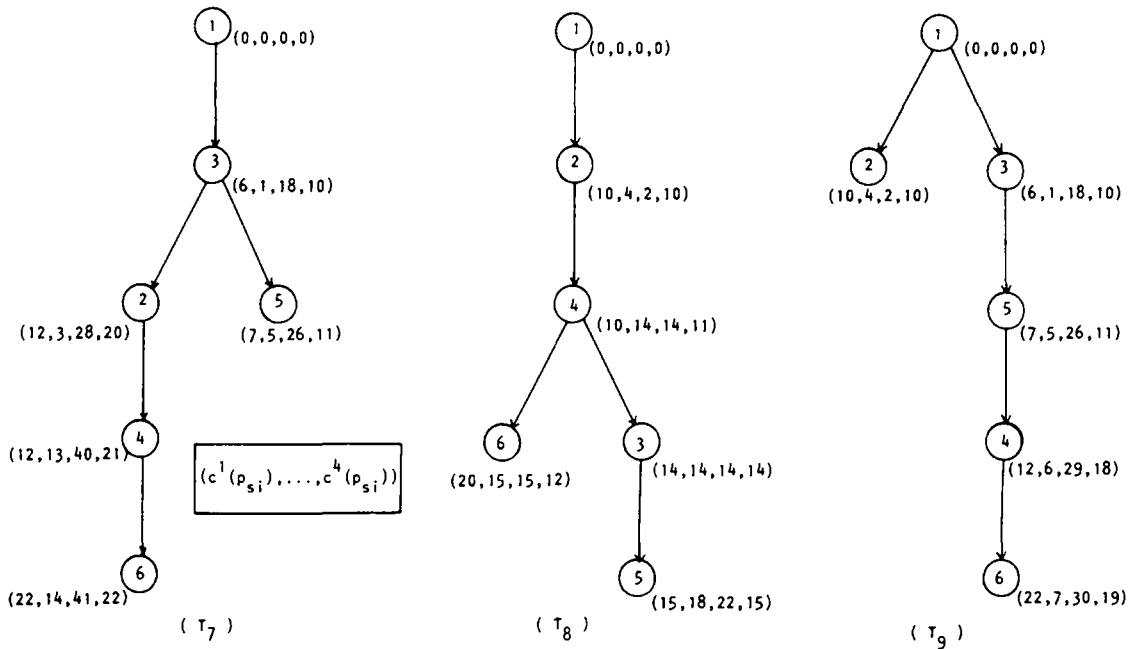


Fig. 7.

not been determined. As the lexicographically shortest tree has been determined, by Lemma 5, there must exist a nondominated path p_{st} adjacent to p'_{st} , for which either $[c^1(p_{st}), \dots, c^r(p_{st})] <^L [c^1(p'_{st}), \dots, c^r(p'_{st})]$ holds or $c^k(p_{st}) = c^k(p'_{st})$ holds, for all $k \in \{1, \dots, r\}$.

As the number of nondominated paths is finite, we may assume without loss of generality, that p_{st} has been determined with the algorithm. So, all nondominated trees where p_{st} is determined have been determined too (Step 3). As $[c^1(p_{st}), \dots, c^r(p_{st})] <^L [c^1(p'_{st}), \dots, c^r(p'_{st})]$ or $c^k(p_{st}) = c^k(p'_{st})$ holds, for all $k \in \{1, \dots, r\}$, we may conclude that for some nondominated tree T (where p_{st} is determined) an arc $(i, j) \notin T$ must exist for which one of the conditions of Step 3 holds and $j \in p_{st}$. So, p'_{st} is in the adjacent tree $T' = T \cup$

$\{(i, j)\} - \{(h, j)\}$ where $(h, j) \in T$. As p'_{st} does not belong to LIST1 we may conclude that T' has been removed in Step 4, which is impossible because T' is nondominated. In fact, from T to T' only two branches are different; one corresponding to p'_{st} in T' and p_{sh} (the sub-path of p_{st} in T) that is nondominated.

This ends the proof of the algorithm.

We must remark that the conditions imposed in Step 3 are not sufficient but necessary conditions for an adjacent tree of a nondominated one to be also nondominated. So, some adjacent trees generated in Step 3 can be dominated trees—the reason why Step 4 has been included. A sufficient condition may considerably improve the efficiency of the algorithm, because the execution of Step 4 is a hard task.

Table 5

$(i, j) \notin T_4$	$\bar{c}_{ij}^1$	$\bar{c}_{ij}^2$	$\bar{c}_{ij}^3$	$\bar{c}_{ij}^4$
(3, 2)	2	-1	26	10
(4, 3)	8	13	-4	4
(5, 6)	-7	-10	11	5
(5, 4)	2	-8	15	7

Table 6

$(i, j) \notin T_8$	$\bar{c}_{ij}^1$	$\bar{c}_{ij}^2$	$\bar{c}_{ij}^3$	$\bar{c}_{ij}^4$
(3, 2)	10	12	22	14
(1, 3)	-8	-13	4	-4
(5, 6)	1	3	7	9
(5, 4)	10	5	11	11

8. Conclusion

The presented algorithms have not been implemented in practice. Furthermore, while Algorithm 1 is easily executed, for not very large problems, by hand computation we can not affirm the same for Algorithm 2. In fact, when presenting this algorithm we have insight in its implementation on a computer. Its efficiency depends on the number of nondominated paths, which may become exponential in the number of nodes [5]. Also, the data structure to be used is important. In this case, as all operations are executed over trees we may expect to be successful. In fact, almost all operations can be easily executed if an appropriate data structure is used [4]. The principal problem is the execution of Step 4. Nevertheless, the nondominance test can be executed only for the branches representing paths from s to t . Clearly, more trees are determined but the time saved in Step 4 compensates the effort of generating those additional (dominated) trees. Computational experience will be needed to confirm (or not) this assumption.

As it has been stated before, the ideal would be a sufficient condition to determine nondominated trees adjacent to some predetermined nondominated one.

References

- [1] J.C.N. Climaco and E.Q.V. Martins, On the determination of the nondominated paths in a multiobjective network problem, *Proc. V. Symposium über Operations Research*, Köln, 1980, *Methods in Operations Research* 40 (Anton Hain, Königstein, 1981) 255–258.
- [2] J.C.N. Climaco and E.Q.V. Martins, A bicriterion shortest path algorithm, *European J. Operational Res.* 11 (1982) 399–404.
- [3] W.H. Cunningham, A network simplex method, *Math. Programming* 11 (2) (1976) 105–116.
- [4] B.L. Fox, Data structures and computer science techniques in operations research, *Operations Res.* 26 (5) (1978) 686–717.
- [5] P. Hansen, Bicriterion path problems, in: G. Fandel and T. Gal, Eds., *Multiple Criteria Decision Making: Theory and Applications*, *Lectures Notes in Economics and in Mathematical Systems* 177 (Springer, Heidelberg, 1980) 109–127.
- [6] E.L. Lawler, *Combinatorial Optimization: Networks and Matroids* (Holt Rinehart and Winston, New York, 1976).
- [7] P. Vincke, Problemes multicritères, *Cahiers Centre Etudes Recherche Opérationnelle* 16 (1974) 425–439.
- [8] P.L. Yu and M. Zeleny, The techniques of linear multiobjective programming, *RAIRO* 8 (V-3) (1974) 51–71.
- [9] M. Zeleny, *Linear Multiobjective Programming* (Springer, Berlin, 1974).